Digital Image Acquisition for Tablet Inspection:

* Image Sensor:
    * CCD provides high image quality and low noise
    * CMOS provides high speed and low power consumption
    * Area-scan captures the complete blister pack.
    * High speed cmos is suitable for packaging lines

* Illumination:
    * Backlighting helps identify cracks and missing tablets
    * Diffuse dome lighting reduces reflections from shiny foil.
    * Polarizing filters help reduce glare.

* Optical Components:
    * Lens focuses light onto the sensor
    * Focal length controls magnification and field of view.
    * Aperture controls the amount of incoming light.

* Spatial Sampling:
    * Determine image resolution
    * Higher resolution helps detect small cracks and chipped edges

* Quantization:
    * Converts img. intensity into digital values

* 8-bit provides 256 intensity levels.

* Image Quality Problems:

  * Motion blur: Reduced using short exposure time

  * glare: Reduced using diffuse lighting.

* Pipeline:

Camera - capture → Illumination → Image formation → Sampling and quantization → Noise removal → Defect detection

2) Contrast Streching and Image Enhancement:

a) Contrast Streching:

$$\text{Formula} \Rightarrow S = \frac{r - r_{min}}{r_{max} - r_{min}} \times 255$$

$$S = \frac{r - 60}{100 - 60} \times 255$$

Output:- 0, 64, 128, 191, 255

b) Mean and Dynamic Range:

$$\text{Original mean} = \frac{60+70+80+90+100}{5} = 80$$

$$\text{Enhanced mean} = \frac{0+64+128+191+255}{5} = 127.6$$

Dynamic range = Maximum - Minimum

Image	Mean	Dynamic range
Original	80	40
Enhanced	127.6	255

c) Two Enhancement Operations

1. Median Filtering:

* Removes salt and pepper noise
* Preserves important edges

2. Unsharp Masking:

* Enhances edges and fine details
* Improves visibility of weaving defects

3) Histogram Equalization and Retinal Image Preprocessing

a) Histogram Equalization:

PDF Formula: $PDF = \frac{n_k}{64}$

CDF = Cumulative Sum of PDF

New intensity

$$S_k = \text{round}\left[(L-1) \times CDF\right]$$

$$S_k = \text{round}\left[3 \times CDF\right]$$

r_k	Count	PDF	CDF	New intensity
0	4	0.0625	0.0625	0
1	20	0.3125	0.375	1
2	28	0.4375	0.8125	2
3	12	0.1875	1.000	3

b) Retinal Image Preprocessing:

Dataset: 8,000 retinal image.

Preprocessing steps:

1. Illumination Correction: Background estimation removes uneven brightness

2. Noise Removal: Median filtering removes unwanted noise

3. Contrast Enhancement: CLAHE improves visibility of blood vessels and abnormalities

4. Resizing: Bilinear interpolation resizes images to 224 × 224 pixels

5. Normalization: Converts pixels values from 0-255 into 0

4) Image Smoothening and Sharpening in PCB Inspection

1) Weighted Averaging:

Gaussian approximation kernel:

$$K = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

Formula:

$$I_S = \frac{\Sigma (K \times P)}{16}$$

It reduces noise and smooths the image.

2) Sharpening:

$$K_S = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Formula:

$$I_{sharp} = 9P_{center} - \Sigma P_{neighbours}$$

The result is limited to 0-255

3. Difference:

$$Difference = |I_{original} - I_S|$$

4) comparision:

Smoothing	Sharpening
Reduces noise	Enhances edges
Produces smooth image	Produces sharp image
Removes small variations	Highlights defects

5) Image Stitching for Autonomous Drone

Introduction: Image stitching combines overlapping aerial images into one continuous farm map.

1) Canny Edge Detection:

* Detects important edges
* Identifies roads and field boundaries
* Reduces unnecessary structures

2) Hough Transform:

* Detects straight lines
* Identifies roads and field boundaries
* Helps analyze geometric structures.

3) ORB / SIFT Keypoint Matching :

* Detects important image features
* SIFT handles rotation and scale changes
* ORB is faster and suitable for real

4) RANSAC Homography:

* Removes incorrect feature matches
* Estimates the homography transformation.
* Aligns overlapping images accurately.

Homography:

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

5) Image Warping and Blending.

* Warping aligns images into a common coordinate system.
* Blending system overlapping regions
* Brightness correction reduces lighting differences

6) Proposed Pipeline:

Input aerial image
↓
Canny edge detection
↓
SIFT keypoint extraction
↓
Feature matching
↓
RANSAC homography
↓
Image warping and blending